Practice Quiz 07A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 07

Transitivity and equivalence classes of congruence.

Recall that $u \equiv v \pmod 3$ means $3 \mid (u - v)$, and that $d \mid n$ means $n = dk$ for some integer k.

1. Define a relation on $\mathbf{Z}$ by $x \sim y$ if and only if $x \equiv y \pmod 3$.

   (a) Prove directly from the definition that $\sim$ is transitive.

   Solution Let $x, y, z \in \mathbf{Z}$ be arbitrary and suppose $x \sim y$ and $y \sim z$. Then $3 \mid (x - y)$ and $3 \mid (y - z)$, so there are integers j, k such that

   $$x - y = 3j \qquad \text{and} \qquad y - z = 3k.$$

   Adding gives $x - z = (x - y) + (y - z) = 3(j + k)$. Because $j + k$ is an integer, $3 \mid (x - z)$. Hence $x \equiv z \pmod 3$, so $x \sim z$. Therefore $\sim$ is transitive. ∎

   (b) Assume that $\sim$ is also reflexive and symmetric. State the number of equivalence classes and list the classes.

   Solution There are exactly three classes:

   $$[0] = \{3k : k \in \mathbf{Z}\}, \qquad [1] = \{3k + 1 : k \in \mathbf{Z}\}, \qquad [2] = \{3k + 2 : k \in \mathbf{Z}\}.$$

   Every integer has exactly one remainder among $0, 1, 2$ on division by 3. Integers are related exactly when they have the same remainder, so these are all the classes and no two are equal. ∎

Solutions revised using GPT-6 Astra in the Codex harness.